

3.1.2. *Results for the L_ψ general affine surface areas.* In the case of the L_ψ general affine surface areas, we deduce the following results. Their proofs follow in completely analogous fashion to those for the L_ϕ general surface areas in the results above, and hence are omitted.

Theorem 3.6. *Let $\psi \in \text{Conv}(0, \infty)$ and $a, b \geq 0$, with a, b not both zero. Suppose that the function $G(t) = \psi(t^{n+1})$ is convex for $t \in (0, \infty)$. Let $K_1, K_2 \in \mathcal{K}_o^n$ be such that $K_1|_{e_n^\perp} = K_2|_{e_n^\perp}$, and assume that K_1, K_2 have positive Gaussian curvature almost everywhere. Then*

$$\text{as}_\psi(a \diamond_{e_n} K_1 \oplus_{e_n} b \diamond_{e_n} K_2) \leq \frac{a}{a+b} \cdot \text{as}_\psi((a+b) \diamond_{e_n} K_1) + \frac{b}{a+b} \cdot \text{as}_\psi((a+b) \diamond_{e_n} K_2).$$

In the special case $a+b=1$, we obtain the following

Corollary 3.3. *Let $\psi \in \text{Conv}(0, \infty)$ and $\lambda \in [0, 1]$ be given. Suppose that the function $G(t) = \psi(t^{n+1})$ is convex for $t \in (0, \infty)$. Let $K_1, K_2 \in \mathcal{K}_o^n$ be such that $K_1|_{e_n^\perp} = K_2|_{e_n^\perp}$, and assume that K_1, K_2 have positive Gaussian curvature almost everywhere. Then we have*

$$\text{as}_\psi(\lambda \diamond_{e_n} K_1 \oplus_{e_n} (1-\lambda) \diamond_{e_n} K_2) \leq \lambda \text{as}_\psi(K_1) + (1-\lambda) \text{as}_\psi(K_2).$$

Remark 3.4. *Let $K \in \mathcal{K}_o^n$ and $\psi \in \text{Conv}(0, \infty)$, and suppose that the function $G(t) = \psi(t^{n+1})$ is convex for $t \in (0, \infty)$. Ye [49, Theorem 3.10] proved that if K has positive Gaussian curvature almost everywhere, then*

$$\text{as}_\psi(K) \geq \text{as}_\psi(S_{e_n}(K)).$$

Again recalling that $\frac{1}{2} \diamond_{e_n} K \oplus_{e_n} \frac{1}{2} \diamond_{e_n} R_{e_n}(K_1) = S_{e_n}(K)$, we see that Corollary 3.3 extends [49, Theorem 3.10] from the case of the Steiner symmetral of a single convex body to the graph sum of two convex bodies.

Corollary 3.4. *Let $\psi \in \text{Conv}(0, \infty)$ be given. Let $K_1, K_2 \in \mathcal{K}_o^n$ be convex bodies with curvature functions and which satisfy $K_1|_{e_n^\perp} = K_2|_{e_n^\perp}$. If the function $G(t) = \psi(t^{n+1})$ is convex for $t \in (0, \infty)$, then*

$$\lambda \text{as}_\psi^*(K_1) + (1-\lambda) \text{as}_\psi^*(K_2) \geq \text{as}_\psi^*([\lambda \diamond_{e_n} K_1^\circ \oplus_{e_n} (1-\lambda) \diamond_{e_n} K_2^\circ]^\circ).$$

Remark 3.5. *Let $\psi \in \text{Conv}(0, \infty)$ and suppose that $G(t) = \psi(t^{n+1})$ is convex for $t \in (0, \infty)$. Ye [49, Corollary 3.11] proved that for every $K \in \mathcal{K}_o^n$ with a curvature function and for which $[S_{e_n}(K^\circ)]^\circ$ has a curvature function, we have*

$$\text{as}_\psi^*(K) \geq \text{as}_\psi^*([S_{e_n}(K^\circ)]^\circ).$$

Thus, in the special case $K_2 = R_{e_n}(K_1)$, Corollary 3.4 reduces to [49, Corollary 3.11].

3.2. Minkowski's first inequality for the L_p affine surface area of a graph sum. For $K \in \mathcal{K}_o^n$ and $p \neq -n$, the L_p affine surface area $\text{as}_p(K)$ is defined by

$$(3.22) \quad \text{as}_p(K) = \int_{\partial K} \frac{\kappa_K(x)^{\frac{p}{n+p}}}{\langle x, N_K(x) \rangle^{\frac{n(p-1)}{n+p}}} d\mu_{\partial K}(x),$$

provided the integral exists. The L_p affine surface area is a key affine invariant in the L_p Brunn–Minkowski theory. It was introduced by Lutwak [30] for $p > 1$, and by Schütt and